

Letters and scripts

Each expression appears at normal size, in a superscript and in a nested superscript.

01 Latin letters and digits

$$\begin{array}{lll} A, B, C, \dots, X, Y, Z & a, b, c, \dots, x, y, z & 0, 1, 2, \dots, 9 \\ \mathcal{A}, \mathcal{B}, \mathcal{C}, \dots, \mathcal{X}, \mathcal{Y}, \mathcal{Z} & \mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \dots, \mathfrak{x}, \mathfrak{y}, \mathfrak{z} & \mathbf{0}, \mathbf{1}, \mathbf{2}, \dots, \mathbf{9} \\ {}^2\mathcal{A}, {}^2\mathcal{B}, {}^2\mathcal{C}, \dots, {}^2\mathcal{X}, {}^2\mathcal{Y}, {}^2\mathcal{Z} & {}^2\mathfrak{a}, {}^2\mathfrak{b}, {}^2\mathfrak{c}, \dots, {}^2\mathfrak{x}, {}^2\mathfrak{y}, {}^2\mathfrak{z} & {}^2\mathbf{0}, {}^2\mathbf{1}, {}^2\mathbf{2}, \dots, {}^2\mathbf{9} \end{array}$$

02 Greek letters

$$\begin{array}{ll} \Gamma, \Delta, \Theta, \Lambda, \Xi, \Pi, \Sigma, \Phi, \Psi, \Omega & \alpha, \beta, \gamma, \delta, \varepsilon, \theta, \varphi, \omega \\ {}^2\Gamma, {}^2\Delta, {}^2\Theta, {}^2\Lambda, {}^2\Xi, {}^2\Pi, {}^2\Sigma, {}^2\Phi, {}^2\Psi, {}^2\Omega & {}^2\alpha, {}^2\beta, {}^2\gamma, {}^2\delta, {}^2\varepsilon, {}^2\theta, {}^2\varphi, {}^2\omega \\ {}^2{}^2\Gamma, {}^2{}^2\Delta, {}^2{}^2\Theta, {}^2{}^2\Lambda, {}^2{}^2\Xi, {}^2{}^2\Pi, {}^2{}^2\Sigma, {}^2{}^2\Phi, {}^2{}^2\Psi, {}^2{}^2\Omega & {}^2{}^2\alpha, {}^2{}^2\beta, {}^2{}^2\gamma, {}^2{}^2\delta, {}^2{}^2\varepsilon, {}^2{}^2\theta, {}^2{}^2\varphi, {}^2{}^2\omega \end{array}$$

03 Cardinals and relations

$$\begin{array}{ll} \aleph_\alpha \times \aleph_\beta = \aleph_\beta & \iff \alpha \leq \beta \\ {}^2\aleph_\alpha \times {}^2\aleph_\beta = {}^2\aleph_\beta & \iff \alpha \leq \beta \\ {}^2{}^2\aleph_\alpha \times {}^2{}^2\aleph_\beta = {}^2{}^2\aleph_\beta & \iff \alpha \leq \beta \end{array}$$

04 Limits and implications

$$\begin{array}{ll} |x - a| < \delta & \implies |f(x) - L| < \varepsilon \\ {}^2|x - a| < \delta & \implies |f(x) - L| < \varepsilon \\ {}^2{}^2|x - a| < \delta & \implies |f(x) - L| < \varepsilon \end{array}$$

05 Sets and topology

$$\begin{array}{ll} \{x \mid x \neq x\} = \emptyset & (A \cap B)^\circ \subseteq A^\circ \cap B^\circ \\ {}^2\{x \mid x \neq x\} = \emptyset & (A \cap B)^\circ \subseteq A^\circ \cap B^\circ \\ {}^2{}^2\{x \mid x \neq x\} = \emptyset & (A \cap B)^\circ \subseteq A^\circ \cap B^\circ \end{array}$$

06 Tensor indices

$$\begin{array}{l} R_{ijkl} = -R_{jikl} = -R_{ijlk} = R_{klij} \\ {}^2R_{ijkl} = -{}^2R_{jikl} = -{}^2R_{ijlk} = {}^2R_{klij} \\ {}^2{}^2R_{ijkl} = -{}^2{}^2R_{jikl} = -{}^2{}^2R_{ijlk} = {}^2{}^2R_{klij} \end{array}$$

Calculus and compound expressions

Each expression appears at normal size, in a superscript and in a nested superscript.

07 Composition and derivatives

$$(f \circ g)'(x) = f'(g(x)) g'(x)$$

$$2^{(f \circ g)'(x) = f'(g(x)) g'(x)}$$

$$2^2^{(f \circ g)'(x) = f'(g(x)) g'(x)}$$

08 Differential forms

$$d\omega = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx \wedge dy$$

$$2^{d\omega = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx \wedge dy}$$

$$2^2^{d\omega = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx \wedge dy}$$

09 Vector fields

$$X = \sum_i \xi^i \frac{\partial}{\partial x^i} + \sum_j \eta^j \frac{\partial}{\partial \dot{x}^j}$$

$$2^{X = \sum_i \xi^i \frac{\partial}{\partial x^i} + \sum_j \eta^j \frac{\partial}{\partial \dot{x}^j}}$$

$$2^2^{X = \sum_i \xi^i \frac{\partial}{\partial x^i} + \sum_j \eta^j \frac{\partial}{\partial \dot{x}^j}}$$

10 Gaussian integral

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$

$$2^{\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}}$$

$$2^2^{\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}}$$

11 Piecewise expressions

$$f(x) = \begin{cases} x^2, & x \geq 0, \\ -x^2, & x < 0. \end{cases}$$

$$2^{f(x) = \begin{cases} x^2, & x \geq 0, \\ -x^2, & x < 0. \end{cases}}$$

$$2^2^{f(x) = \begin{cases} x^2, & x \geq 0, \\ -x^2, & x < 0. \end{cases}}$$

12 Products and annotations

$$\underbrace{V \times \cdots \times V}_k \times \underbrace{V \times \cdots \times V}_l = V^{k+l}$$

$$2^{\underbrace{V \times \cdots \times V}_k \times \underbrace{V \times \cdots \times V}_l = V^{k+l}}$$

$$2^2^{\underbrace{V \times \cdots \times V}_k \times \underbrace{V \times \cdots \times V}_l = V^{k+l}}$$

Styles, spacing and accents

13 Math alphabets

$ABCxyz$ $\mathbf{ABCxyz}$ $\mathcal{ABCxyz}$ $\mathbb{C}\mathbb{H}\mathbb{N}\mathbb{R}\mathbb{Z}$

14 Primes

f' f' f'' f'' f''' f'''
 x'_i x'_i $f'(x) + g''(x) + h'''(x)$

15 Ordinary spacing and scripts

$f1$ $f0$ fA $f\alpha$ $V1$ $W1$ $j1$ ff fi
 f_i f^2 f_i^2 $X_f + X_j + X_p + X_y + X_A + X_M$

16 Single-character accents

$\hat{x}$ $\tilde{y}$ $\check{z}$ $\bar{M}$ $\vec{A}$ $\dot{\Gamma}$ $\ddot{\Gamma}$

17 Wide accents

$\widehat{xy}$ $\widehat{xyz}$ $\widehat{x + y + z + w}$
 $\widetilde{xy}$ $\widetilde{xyz}$ $\widetilde{x + y + z + w}$

18 Bars and overlines

$\bar{M}$ $\overline{xy}$ $\overline{x + y + z + w}$

19 Radicals

$\sqrt{x}$ $\sqrt{x^2 + y^2}$ $\sqrt[n]{\frac{a}{b}}$ $\sqrt{\frac{1 + \sqrt{1 + x^2}}{2}}$

Braces, delimiters and operators

20 Overbraces

$$\overbrace{A} \quad \overbrace{ABCDEF} \quad \overbrace{ABCDEFGH I J K L M N}$$

21 Underbraces

$$\underbrace{A} \quad \underbrace{ABCDEF} \quad \underbrace{ABCDEFGH I J K L M N}$$

22 Brace annotations

$$\overbrace{a_1 + a_2 + \cdots + a_n}^{n \text{ terms}} \quad \underbrace{x_1 x_2 \cdots x_m}_{m \text{ factors}}$$

23 Tall delimiters

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

24 Nested fractions and delimiters

$$\left\{ \frac{1}{1 + \frac{1}{1+x}} \right\} \quad \left\langle \frac{u+v}{\sqrt{1+\|u\|^2}}, w \right\rangle$$

25 Limits and large operators

$$\sum_{k=0}^n \binom{n}{k} x^k y^{n-k} = (x+y)^n$$

$$\int_0^1 \frac{dx}{\sqrt{1-x^2}} = \frac{\pi}{2} \quad \prod_{k=1}^n \frac{k+1}{k} = n+1$$

26 Composite expressions

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

$$R_{\sigma\mu\nu}^{\rho} = \partial_{\mu} \Gamma_{\nu\sigma}^{\rho} - \partial_{\nu} \Gamma_{\mu\sigma}^{\rho} + \Gamma_{\mu\lambda}^{\rho} \Gamma_{\nu\sigma}^{\lambda} - \Gamma_{\nu\lambda}^{\rho} \Gamma_{\mu\sigma}^{\lambda}$$

$$\Gamma_{\mu\nu}^{\rho} = \frac{1}{2} g^{\rho\sigma} \left(\frac{\partial g_{\sigma\nu}}{\partial x^{\mu}} + \frac{\partial g_{\sigma\mu}}{\partial x^{\nu}} - \frac{\partial g_{\mu\nu}}{\partial x^{\sigma}} \right)$$

$$\int_{\partial M} \omega = \int_M d\omega \quad \iint_S (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} dS = \oint_{\partial S} \mathbf{F} \cdot d\mathbf{r}$$

$$|jm\rangle = \sum_{m_1=-j_1}^{j_1} \sum_{m_2=-j_2}^{j_2} (-1)^{j_1-j_2+m} \sqrt{2j+1} \begin{pmatrix} j_1 & j_2 & j \\ m_1 & m_2 & -m \end{pmatrix} |j_1 m_1\rangle \otimes |j_2 m_2\rangle$$